

INSTRUCTFX2FX: A MULTI-TURN TEXT-TO-PRESET DEMO FOR ITERATIVE AUDIO EFFECT REFINEMENT

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ABSTRACT

We present *InstructFX2FX*, an interactive demonstration of multi-turn, text-guided audio effect refinement. Existing text-to-preset systems are largely single-shot, mapping one textual descriptor to one preset. Real audio engineering is instead sequential: engineers refine an existing effect chain through successive instructions such as “make it warmer” or “too harsh, soften it.” This poses a stateful problem that single-shot systems do not address — given the current effect parameters and a new instruction, update the sound while preserving what earlier instructions already achieved. *InstructFX2FX* tackles this with a hybrid architecture that divides labor between a language model and CLAP-guided optimization. The LLM acts as a high-level planner that selects effects, orders the signal chain, and proposes the initial parameter state, leveraging recent evidence that LLMs can outperform CLAP-based optimization for single-turn text-to-parameter mapping [1]; CLAP-guided optimization then refines the existing parameter state from this initialization, avoiding the parameter drift caused by re-prompting the LLM at each turn. At the demo, attendees drive a dry recording through successive natural-language instructions while staying in the loop: after each turn they audition the saved optimization checkpoints to choose how strongly an effect is applied, then issue the next instruction based on what still differs from the sound they intend. On SocialFX-derived descriptor transitions, CLAP-guided refinement lowers target-directed DSP-feature MMD on 9 of 10 directed descriptor pairs relative to an LLM-only re-prompting baseline, with most refinement trajectories trending toward the intended target. Audio demos are available, and the code will be open-sourced, at <https://instructfx2fx.vaclis.net> and <https://github.com/vaclisinc/InstructFX2FX>.

1. INTRODUCTION

Audio effect design is rarely a one-shot process. In practice, audio engineers and producers establish a rough effect chain and then iteratively refine it through feedback such as “make it brighter,” “add some weight,” or “ease off the reverb.” Each instruction is interpreted relative to the current sound and parameter state, not as an independent request from scratch.

We call this problem *sequential FX refinement*: each new instruction should update the existing effect parameters so that the rendered audio tracks the user’s evolving intent while preserving

relevant structure from the current state. This differs from conventional text-to-preset generation, where each prompt is processed independently. We formalize this distinction in Section 2.

Recent works address only the single-turn case. Text2FX and FxSearcher both optimize effect parameters against a CLAP objective, differing mainly in the optimizer: Text2FX uses gradient descent through differentiable effects from a random initialization [2], while FxSearcher uses gradient-free Bayesian optimization for non-differentiable plugins [3]. LLM-based systems instead map text directly to parameters: LLM2Fx claims that an LLM, drawing on pretrained knowledge and in-context examples, can outperform CLAP-based optimization [1], and a concurrent tool-calling variant further selects and orders the FX chain through chain-of-thought planning [4]; both, however, remain single-shot. None of these directly support sequential refinement, and naively re-prompting an LLM at each turn is unreliable: the model regenerates parameters non-deterministically, perturbs settings the user did not ask to change, and — most fundamentally — cannot hear the sound it is editing.

We therefore propose a hybrid design. The language model acts as the “brain” of the system: it selects which effects to apply, orders the signal chain, and generates reasonable initial parameters. CLAP-guided optimization then acts as the “hand” and “ear” of the system: it adjusts the effect parameters and listens to the rendered result through CLAP, iteratively pulling the audio embedding toward the user instruction in CLAP space. In this view, the LLM provides the starting point, while CLAP provides the refinement direction.

We realize these ideas in *InstructFX2FX*, a session-aware demonstration system. This demo paper makes three contributions. First, we formulate sequential FX refinement as a stateful, multi-turn parameter-update problem, distinct from single-shot text-to-preset generation. Second, we build a system that routes each instruction to reuse, initialize, or jointly refine an effect chain, pairing LLM-based planning with CLAP-guided gradient descent and Bayesian optimization. Third, we provide an interactive demonstration together with a preliminary evaluation on SocialFX descriptor transitions.

2. PROBLEM FORMULATION

We formulate sequential FX refinement as a stateful parameter-update problem. Let x denote the dry input audio, C_t the current effect chain at turn t , and P_t the corresponding effect parameters. Given a user instruction I_t , the system produces an updated chain and parameter state:

$$(C_t, P_t) = f(x, C_{t-1}, P_{t-1}, I_t, H_{t-1}),$$

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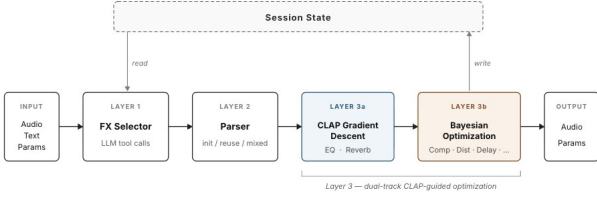


Figure 1: The InstructFX2FX harness. The LLM planner selects and orders the FX chain, the routing module determines how the current session state should be updated, and the optimization backend refines parameters in CLAP space. The persistent session state makes the system iterative.

where $H_{t-1} = \{I_1, \dots, I_{t-1}\}$ denotes the prior instruction history.

This formulation differs from single-shot text-to-preset generation:

$$P = g(x, I),$$

where each prompt is processed independently. In sequential refinement, the desired behavior is not merely to generate a plausible preset for the latest instruction, but to update the existing sound in a controlled direction. For example, after the sequence “make it warm” → “make it heavier,” the second instruction should preserve the warmth already introduced while moving the sound toward heaviness. Meeting this goal requires persistent state across turns, a decision about whether each instruction reuses or extends the chain, and parameter refinement driven by the actual rendered audio — the three design choices realized in Section 3.

3. SYSTEM OVERVIEW

Figure 1 shows the InstructFX2FX architecture: an LLM planner, a routing module, and a CLAP-guided optimization backend, all operating over a persistent session state that stores the current effect chain, parameter values, rendered checkpoints, and instruction history. As outlined above, the LLM makes the symbolic decisions while CLAP-guided optimization supplies the perceptual feedback it lacks; the rest of this section details how each instruction is routed (Section 3.1) and how its parameters are optimized (Section 3.2).

3.1. Session-aware Multi-turn Refinement

A central requirement of sequential FX refinement is that each instruction should operate on the existing state. InstructFX2FX therefore maintains a persistent session state across prompts and reads it at every turn before deciding how to respond.

The routing module dispatches each instruction into one of three execution modes:

- **Initialize-only:** no existing effects are present, so the LLM initializes a new chain and parameters.
- **Reuse-and-optimize:** all relevant effects already exist, so the system refines the current parameters in place.
- **Mixed reuse-and-initialize:** part of the existing chain is reused, while newly required effects are initialized and inserted.

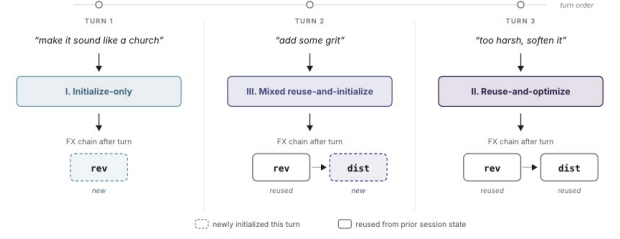


Figure 2: The three routing modes used in multi-turn refinement. Initialize-only creates a new chain, mixed reuse-and-initialize extends the existing chain, and reuse-and-optimize refines the current chain in place.

Using Figure 2 as an example, the instruction sequence “make it sound like a church” → “add some grit” → “too harsh, soften it” triggers three different behaviors. The first turn initializes a reverb effect, the second adds distortion on top of the existing reverb chain, and the third performs corrective refinement on the current chain without introducing additional effects. Corrective instructions are thus treated as local updates to the current sound, while additive instructions extend the chain when needed.

3.2. Optimization Backends

After the LLM planner determines the effect chain and initial parameters, InstructFX2FX refines the selected parameters using existing CLAP-guided optimization objectives. Our contribution is not a new optimizer, but a session-aware framework that integrates these optimization methods into an interactive multi-turn workflow. Table 1 shows the effects currently supported in the demonstration system and their corresponding optimization backends.

Both backends optimize CLAP-guided objectives adopted from prior text-guided FX work, by gradient descent for differentiable effects and by Bayesian search for non-differentiable ones. Let $\phi_{\text{audio}}(\cdot)$ and $\phi_{\text{text}}(\cdot)$ denote the CLAP audio and text embedding functions [5].

The semantic similarity objective directly aligns rendered audio with the target instruction:

$$\mathcal{L}_{\text{sem}} = 1 - \cos(\phi_{\text{audio}}(\mathbf{x}), \phi_{\text{text}}(I)). \quad (1)$$

For sequential refinement, we additionally use a directional objective that encourages the refinement trajectory in audio-embedding space to follow the semantic transition from a source descriptor A to a target descriptor B :

$$\mathcal{L}_{\text{dir}} = 1 - \cos(\phi_{\text{audio}}(\mathbf{x}(t)) - \phi_{\text{audio}}(\mathbf{x}_0), \phi_{\text{text}}(B) - \phi_{\text{text}}(A)). \quad (2)$$

We also support a guided objective with a positive target I^+ and a negative anchor I^- , following prior CLAP-guided optimization work:

$$\mathcal{L}_{\text{guided}} = 1 - \cos(\phi_{\text{audio}}(\mathbf{x}), \phi_{\text{text}}(I^+)) + \cos(\phi_{\text{audio}}(\mathbf{x}), \phi_{\text{text}}(I^-)). \quad (3)$$

In both cases, optimization starts from the LLM-generated or session-reused parameter state. Because the desired strength of an effect is subjective, the system saves snapshots along the optimization trajectory and exposes them in the demo interface as a slider, letting the user audition intermediate results and choose how strong an effect they want rather than committing to only the final render.

Table 1: Effects currently supported in the demonstration system.

Category	Effects	Optimization
Diff.	EQ, reverb	Gradient descent
Non-diff.	Compressor, distortion, delay, bitcrush, pitch shift	Bayesian optimization

4. EVALUATION

We run a preliminary evaluation on EQ descriptor transitions from the SocialFX dataset [6], testing whether CLAP-guided refinement improves on both an LLM-only reprompting baseline and the LLM’s own initialization.

4.1. Dataset and Ground Truth

We evaluate on the SocialFX dataset, selecting seven EQ descriptor words by frequency (≥ 20 occurrences) and consistency (≥ 0.45): *warm* (93), *bright* (74), *soft* (74), *loud* (54), *harsh* (39), *calm* (26), and *heavy* (25). From these we construct 10 directed word pairs (e.g., *warm* $\rightarrow$ *bright*, *harsh* $\rightarrow$ *soft*).

Ground truth audio is produced sequentially: for a pair $A \rightarrow B$, we apply the SocialFX EQ parameters for word A to dry audio, then apply the parameters for word B . For each word and instrument condition (piano and violin), we sample 50 random parameter instances.

4.2. Metric: Maximum Mean Discrepancy

All comparisons use Maximum Mean Discrepancy (MMD) with a Gaussian kernel (median bandwidth heuristic) over 35-dimensional DSP feature vectors (spectral centroid, bandwidth, rolloff, MFCCs, temporal features). Lower MMD indicates that the system output distribution is closer to the sequential ground truth distribution.

This metric is intentionally separate from the CLAP objective used during optimization, so an improvement in MMD indicates that CLAP-guided refinement transfers to a DSP-feature-based evaluation.

4.3. Experiment 1: Sequential MMD

We compare InstructFX2FX against an LLM-only baseline (LLM reprompting without CLAP optimization) on all 10 directed word pairs. As shown in Figure 3a, our system achieves lower MMD on **9 of 10** pairs: CLAP gradient descent consistently outperforms LLM-only reprompting for these EQ descriptors. The largest gains occur on pairs with clear timbral contrast (e.g., *heavy* $\rightarrow$ *calm*, *harsh* $\rightarrow$ *soft*); the smallest gains are on pairs where the CLAP objective and DSP-feature MMD are less aligned (e.g., *calm* $\rightarrow$ *loud*).

4.4. Experiment 2: MMD Convergence Trajectory

The convergence trajectories in Figure 4 show that CLAP-guided refinement does not always improve the sound monotonically. Because lower MMD means the rendered audio is closer to the sequential ground truth, a healthy trajectory should fall and stay below the LLM-only baseline. For *loud* $\rightarrow$ *heavy*, both instruments do exactly this throughout optimization. For *calm* $\rightarrow$ *loud*,

however, the violin first drifts *above* the baseline — briefly moving *away* from the target — for roughly the first 400 iterations before recovering. This recurring failure mode, in which early CLAP progress overshoots the DSP-feature optimum, exposes a mismatch between the CLAP optimization objective and the MMD evaluation metric.

4.5. Experiment 3: LLM Initialization Ablation

Unlike the sequential pairs above, this ablation targets eight single descriptor words (the seven plus *happy*): starting from the same LLM-initialized parameters, we compare final DSP-feature MMD with and without CLAP gradient descent. Adding CLAP refinement lowers the overall MMD from 0.4461 to 0.3380 (about **24%**) and surpasses the initialization in CLAP space for **7 of 8** words (Figure 3b). The LLM-only condition is flat across iterations, confirming that CLAP gradient descent provides a reliable refinement signal rather than merely a better starting point.

5. LIMITATIONS AND FUTURE WORK

Several optimization and evaluation challenges remain. First, our quantitative evaluation centers primarily on EQ-related descriptors, because SocialFX provides cleaner descriptor-to-parameter associations for EQ than for more complex effects.

Second, CLAP embedding space does not capture all perceptual properties relevant to audio production [7]. Although CLAP-guided refinement often improves early behavior, trajectories may later drift or overshoot, and gains in CLAP space do not always correspond to gains in DSP-feature metrics or perceptual quality.

Third, non-differentiable effects remain difficult to refine reliably. Bayesian optimization extends support to Pedalboard effects such as distortion, delay, and bitcrushing, but its trajectories are often noisy and semantically inconsistent across iterations. Corrective prompts such as “soften it” after distortion are particularly unstable, suggesting that CLAP-based black-box objectives alone may be insufficient for robust sequential refinement.

Finally, optimization currently requires several seconds per interaction turn, preventing real-time deployment inside production DAW workflows.

Future directions include richer multi-effect chains, human listening studies for perceptual validation, domain-adapted audio-effect embedding models, improving stability for non-differentiable refinement, and real-time VST/AU integration.

6. CONCLUSION

We presented InstructFX2FX, a session-aware demonstration system for sequential text-guided audio effect refinement. Unlike single-shot text-to-preset systems, InstructFX2FX treats audio effect editing as a stateful process in which each instruction updates an existing chain and parameter state.

The core idea is a hybrid division of labor: the LLM provides high-level effect planning and initialization, while CLAP-guided optimization provides perceptual refinement. Preliminary experiments on SocialFX-derived descriptor transitions suggest that CLAP-based refinement can improve over LLM-only reprompting in many sequential settings, while also exposing important limitations such as CLAP/DSP-feature mismatch, optimization drift, and instability for non-differentiable effects.

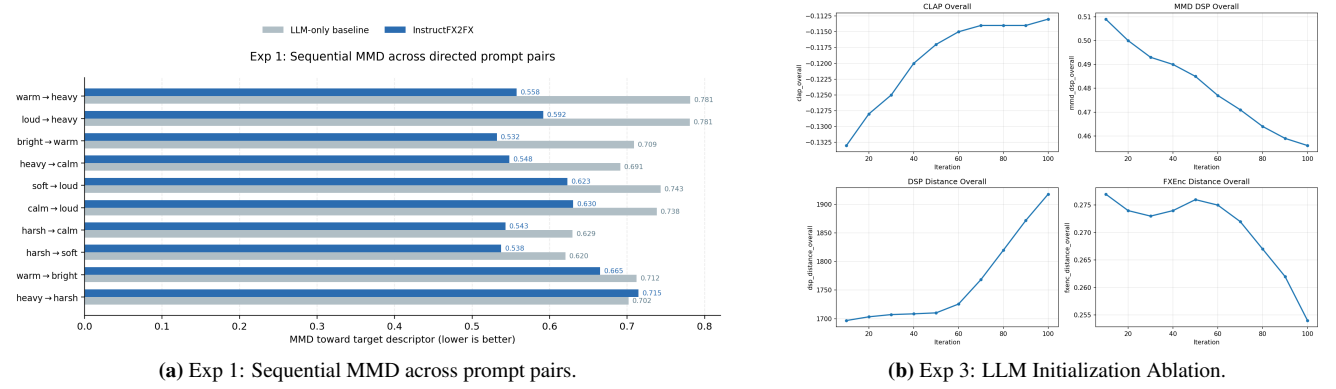


Figure 3: Two summary evaluation results. Left: in Exp 1, InstructFX2FX improves target-directed MMD on most directed prompt pairs relative to an LLM-only baseline. Right: in Exp 3, CLAP refinement improves over the same LLM initialization rather than merely benefiting from a better starting point (see Section 4 for the aggregate MMD reduction).

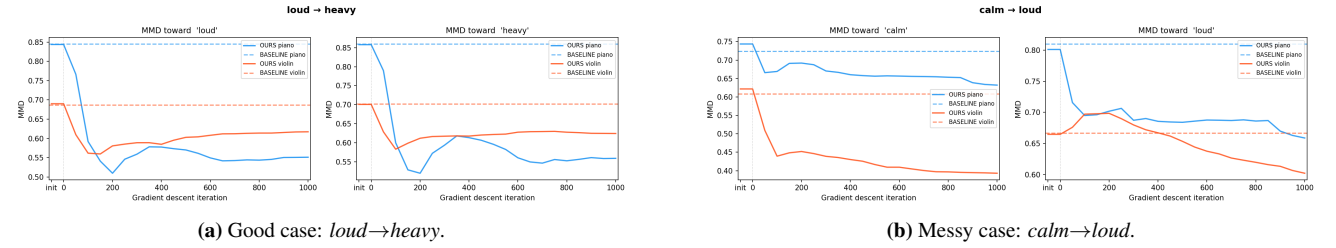


Figure 4: Exp 2: MMD Convergence Trajectory. Some prompt pairs show an early drop in MMD and stay below the baseline, while others overshoot or drift later in optimization, motivating the demo’s intermediate-checkpoint slider.

More broadly, this demo is a proof-of-concept for combining complementary sources of machine intelligence into interactive audio-effect systems that better reflect practical creative workflows. We see it as a first step, and hope it motivates the community to take up sequential FX refinement as a problem in its own right. The most promising next direction, in our view, is to move beyond optimization at inference time toward generative parameter models — for example, flow-matching models trained directly on iterative FX-editing trajectories — that learn sequential refinement end-to-end.

7. REFERENCES

- [1] S. Doh, J. Koo, M. A. Martínez-Ramírez, W.-H. Liao, J. Nam, and Y. Mitsufuji, “Can large language models predict audio effects parameters from natural language?” in *Proc. IEEE Workshop Appl. Signal Process. Audio Acoust. (WASPAA)*, 2025.
- [2] A. Chu, P. O’Reilly, J. Barnett, and B. Pardo, “Text2FX: Harnessing CLAP embeddings for text-guided audio effects,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2025.
- [3] H. Ki, J. Kim, M. Kwon, and J. Kim, “FxSearcher: Gradient-free text-driven audio transformation,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2026.
- [4] S. Doh, J. Koo, M. A. Martínez-Ramírez, W. Choi, W.-H. Liao, Q. Wu, J. Nam, and Y. Mitsufuji, “LLM2Fx-Tools: Tool calling for music post-production,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2026.
- [5] Y. Wu, K. Chen, T. Zhang, Y. Hui, M. Nezhurina, T. Berg-Kirkpatrick, and S. Dubnov, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2023.
- [6] T. Zheng, P. Seetharaman, and B. Pardo, “SocialFX: Studying a crowdsourced folksonomy of audio effects terms,” in *Proceedings of the 24th ACM International Conference on Multimedia*, Amsterdam, The Netherlands, 2016, pp. 1334–1338.
- [7] Q. Deng, B. Pardo, and T. N. Pappas, “Do joint language-audio embeddings encode perceptual timbre semantics?” in *Proc. NeurIPS Workshop on AI for Music*, 2025.